

CM0081 Formal Languages and Automata

§ 8.1 Problems That Computers Cannot Solve

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Preliminaries

Textbook

John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman (2007). Introduction to Automata Theory, Languages, and Computation.

Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Undecidable Problems

Recall

Given $L \subseteq \Sigma^*$ and a word $w \in \Sigma^*$, to recall that to decide whether or not $w \in L$ is a (decision) problem on L .

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Definition (informally)

A problem is **undecidable** iff no algorithm can solve it.

Question

Are there undecidable problems?

Undecidable Problems

Theorem

There are undecidable problems.

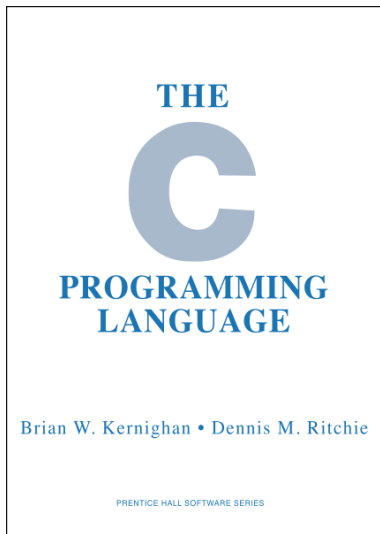
Proof (by cardinality)

Let Σ be an alphabet and let $\overline{\overline{S}}$ be the cardinality of the set S .

- (i) Since Σ is an alphabet, then $\overline{\overline{\Sigma}} = n$, with $n \in \mathbb{Z}^+$, that is, the alphabet Σ is an enumerable set.
- (ii) The set of all the strings over Σ , that is Σ^* , is an enumerable set (lexicographical order).
- (iii) The set of all the languages over Σ , that is, the set $\{L \mid L \subseteq \Sigma^*\}$ is infinite not enumerable set.
- (iv) The set $\{P \mid P \text{ is a program in a general programming language}\}$ is an enumerable set (lexicographical order).

Hence, there are more languages over Σ than programs. In consequence, there must be languages whose decision problems are undecidable. ■

The hello-world Problem: An Undecidable Problem



*“The first program to write is the same for all languages: Print the words **hello, world.**” [1978, §1.1]*

The hello-world Problem: An Undecidable Problem

Problem description

To determine whether a given program, with a given input, prints **hello, world** as the first 12 characters that it prints.

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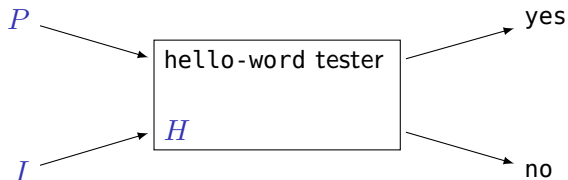
Theorem

The hello-world problem is an undecidable problem.

The hello-world Problem: An Undecidable Problem

Informal proof

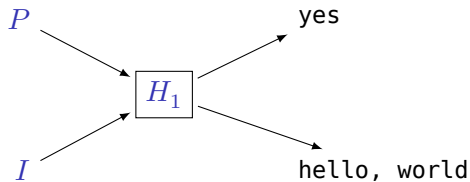
1. We assume that the following program H exists:



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The hello-world Problem: An Undecidable Problem

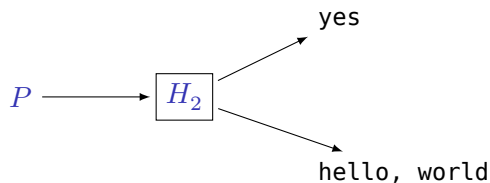
2. From the program H build the program H_1 with the following behaviour:



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The hello-world Problem: An Undecidable Problem

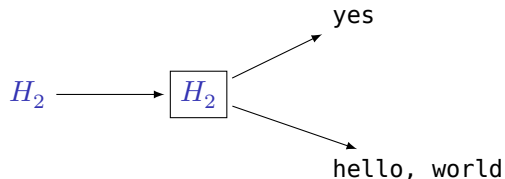
3. To build the program H_2 by restricting H_1 to take only the input P (i.e. P is also the input of the program P):



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The hello-world Problem: An Undecidable Problem

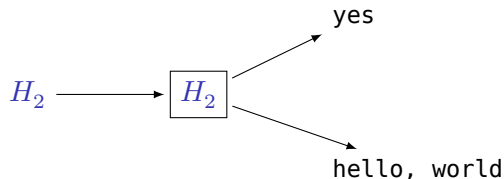
4. What does the program H_2 do when given itself as input?



Analysis on the whiteboard.

The hello-world Problem: An Undecidable Problem

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Analysis on the whiteboard.

Therefore the program H_2 can not exist.

5. Therefore the program H can not exist, i.e. the hello-world problem is undecidable. ■

1900 David Hilbert's problems

Wir müssen wissen — wir werden wissen!

(We must know — we will know!)

Computability: Historical Remarks

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1934-1937 Confluence of ideas

- Derivability from a system of equations (Kurt Gödel, Jacques Herbrand)
- Lambda calculus (Alonzo Church, Stephen C. Kleene, J. Barkley Rosser)
- Recursive functions (Kurt Gödel, Stephen C. Kleene)
- Post machines (Emil L. Post)
- Turing machines (Alan M. Turing)

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Computability: Historical Remarks

1936–40 The Church-Turing-Kleene thesis: A (number-theoretic) function is **effectively calculable** if and only if there is a **Turing machine** which computes it.

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1940–today Many equivalent models of computation

Problems Reduction

How to prove that a problem is undecidable?

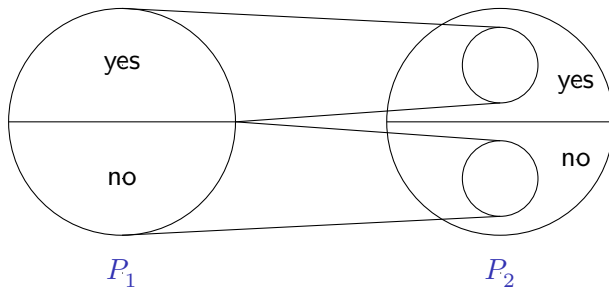
- Proof by contradiction (or constructively speaking, proof of negation (Bauer 2017))

Problems Reduction

How to prove that a problem is undecidable?

- Proof by contradiction (or constructively speaking, proof of negation (Bauer 2017))
- Problem reduction

Reduction from a problem P_1 to a problem P_2 :



Problems Reduction

Theorem

If P_1 is undecidable and there is a reduction of P_1 to P_2 , then P_2 is undecidable too.

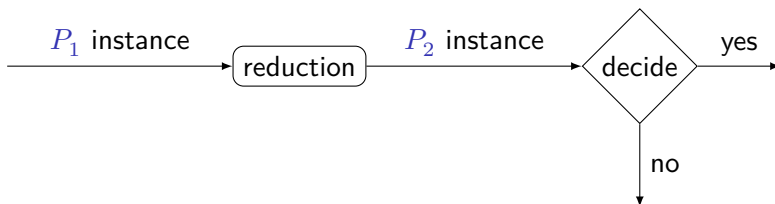
Problems Reduction

Theorem

If P_1 is undecidable and there is a reduction of P_1 to P_2 , then P_2 is undecidable too.

Proof by contradiction

Let's suppose P_2 decidable. The reduction from P_1 to P_2 implies that P_1 is decidable which is a contradiction. Therefore, P_2 is undecidable.



Problems Reduction

Example (the calls-foo problem)

Problem description: Does a program Q with an input z calls the function foo?

Problems Reduction

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Prove that the calls-foo problem is undecidable.

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Problems Reduction

Proof

Idea: Reduction of the `hello-world` problem to the `calls-foo` problem.

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That is, a program P with input y prints **hello, world** iff a program Q with input z calls the function **foo**.

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Reduction:

1. Program P_1 : Rename **foo** in the program P .
2. Program P_2 : Add the function **foo** to the program P_1 .
3. Program P_3 : Save the first 12 characters that prints the program P_2 .
4. Program P_4 : When the program P_3 executes an output statement if output is **hello, world** then calls the function **foo**.
5. $Q = P_4$ and $y = z$. ■

Problems Reduction

Exercise 8.1.1.a (the halting problem)

Problem description: Given a program and an input, does the program eventually halt; i.e. does the program not loop forever on the input?

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Prove that the halting problem is undecidable.

(continued on next slide)

Problems Reduction

Proof (based on the solution in Hopcroft, Motwani and Ullman (2007))

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Idea: Reduction of the **hello-world** problem to the **halting** problem.

That is, a program P with input y prints **hello, world** iff a program Q with input z halts.

Reduction:

1. Program P_1 : Add an infinite loop (e.g. `while(1);`) to the end of `main()`.
2. Program P_2 : Save the first 12 characters that prints the program P_1 .
3. Program P_3 : When the program P_2 executes an output statement if output is **hello, world** then the program P_3 halts by going to the end of `main`.
4. $Q = P_3$ and $y = z$. ■

Problems Reduction

Theorem

If P_1 is undecidable and there is a reduction of P_1 to P_2 , then P_2 is undecidable too.

Be careful

In the above theorem the reduction is from P_1 to P_2 , it is not from P_2 to P_1 :

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$$P_2 \text{ decidable} \Rightarrow P_1 \text{ decidable}$$

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References

- Andrej Bauer (2017). Five States of Accepting Constructive Mathematics. Bulletin of the American Mathematical Society 54.3, pp. 481–498. DOI: [10.1090/bull/1556](https://doi.org/10.1090/bull/1556) (cit. on pp. 20, 21).
- John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on pp. 2, 31–33).